

STRUCTURAL ENGINEERING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an “adequate” score of 7 for an X1 criterion = 7 points; an “adequate” score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Team of two is present
- ☐ The structure is present, identified, and in its storage box
- ☐ The Team Portfolio is present and includes:
 - ☐ The Team Verification form (completed)
 - ☐ Three (3)-view drawing of the structure
 - ☐ A complete cut parts list for the structure
- ☐ ENTRY NOT EVALUATED

PRE-BUILT STRUCTURE (On-Site Destructive Testing) – (40 points)

Indicate N for non-compliant or C for compliant, for each regulation in the Construction section. One non-compliant mark will result in the entry not being evaluated. Dimensional criteria will have a tolerance identified in the design brief.

Regulation	Noncompliant	Compliant
Outside Width of Structure	The outside width of the structure is greater than the designated construction width.	The outside width of the structure is within the designated tolerance of the assigned construction width.
Outside Height of Structure	The height of the structure is greater or less than the designated tolerance of the assigned construction height.	The height of the structure is within the designated tolerance of the assigned construction height.
Inside Structure Width	The inside structure space, if provided in the design brief, is less than the required construction space.	The inside structure space, if provided in the design brief, is greater than the required construction space.
Construction Materials	Material other than the correct construction material was used in the construction of the solution.	Only the correct construction materials are used in the construction of the solution.
Substructure	Substructure, if allowed in the design brief, is not applied correctly.	Substructure, if allowed in the design brief, is applied correctly.
Laminations	Laminations used in the construction of the solution fail to comply with the design brief.	Laminations used in the construction of the solution comply with the design brief.
Coating of Materials	Coating of the construction materials with glue is present.	No coating of the construction materials with glue is present.
Testing Rod and Block Clearance	The testing block and rod cannot be placed and passed through the center of the structure to allow for testing.	The testing block and rod pass freely through the center of the structure to allow for testing.
DISQUALIFIED		
PRE-BUILT STRUCTURE APPROVED FOR TESTING		
Record the mass (weight) of the structure (in grams to the nearest tenth of a gram) prior to testing.		

PRE-BUILT STRUCTURE (On-Site Destructive Testing) – continued					
Record the failure weight in pounds to the nearest tenth of a pound.					
Determine the efficiency (shown to three decimal places) by multiplying the failure weight (or adjusted failure weight, as applicable) by 4.54 and then dividing by the mass (weight) of the structure.					
1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	
6th: 35 Points	7th: 34 Points	8th: 33 Points	9th: 32 Points	10th: 31 Points	
11th: 30 Points	12th: 29 Points	13th: 28 Points	14th: 27 Points	15th: 26 Points	
16th: 25 Points	17th: 24 Points	18th: 23 Points	19th: 22 Points	20th: 21 Points	
PRE-BUILT STRUCTURE TOTAL POINTS (40 points)					

PRELIMINARY SUBTOTAL (40 points)	
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ON-SITE STRUCTURE (Qualification) – (40 points)				
For the ON-SITE STRUCTURE: Indicate N for non-compliant or C for compliant, in the Qualification and Construction sections below. In the Qualification section, one non-compliant mark will result in disqualification. In the Construction section, one non-compliant mark will result in the structure not being evaluated. Dimensional height criteria will have a tolerance identified in the design brief.				
Regulation	Noncompliant		Compliant	
Team of Two	Only one (1) team member is present.		Both team members are present.	
Safety Eyewear	Warnings about eyewear are issued.		No warnings about eyewear are issued.	
Structure Identification	The identification sticker is not attached.		The identification sticker is attached.	
Tools and Fabrication Supplies	Inappropriate tools or supplies are brought to the event.		Appropriate tools and supplies are brought to the event.	
ON-SITE STRUCTURE (Construction)				
Outside Width of Structure	The outside width of the structure is greater than the designated construction width.		The outside width of the structure is within the designated tolerance of the assigned construction width.	
Outside Height of Structure	The height of the structure is greater or less than the designated tolerance of the assigned construction height.		The height of the structure is within the designated tolerance of the assigned construction height.	
Inside Structure Width	The inside structure space, if provided in the design brief, is less than the required construction space.		The inside structure space, if provided in the design brief, is greater than the required construction space.	
Construction Materials	Material other than the correct construction material was used in the construction of the solution.		Only the correct construction materials are used in the construction of the solution.	
Substructure	Substructure, if allowed in the design brief, is not applied correctly.		Substructure, if allowed in the design brief, is applied correctly.	
Laminations	Laminations used in the construction of the solution fail to comply with the design brief.		Laminations used in the construction of the solution comply with the design brief.	
Coating of Materials	Coating of the construction materials with glue is present.		No coating of the construction materials with glue is present.	
Testing Rod and Block Clearance	The testing block and rod cannot be placed and passed through the center of the structure to allow for testing.		The testing block and rod pass freely through the center of the structure to allow for testing.	

ON-SITE STRUCTURE (40 points) – continued					
DISQUALIFIED					
ON-SITE STRUCTURE APPROVED FOR TESTING					
Record the mass (weight) of the structure (in grams to the nearest tenth of a gram) prior to testing.					
Record the failure weight in pounds to the nearest tenth of a pound.					
Determine the efficiency (shown to three decimal places) by multiplying the failure weight (or adjusted failure weight, as applicable) by 4.54 and then dividing by the mass (weight) of the structure.					
1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	
6th: 35 Points	7th: 34 Points	8th: 33 Points	9th: 32 Points	10th: 31 Points	
11th: 30 Points	12th: 29 Points	13th: 28 Points	14th: 27 Points	15th: 26 Points	
16th: 25 Points	17th: 24 Points	18th: 23 Points	19th: 22 Points	20th: 21 Points	
ON-SITE STRUCTURE TOTAL POINTS (40 points)					

SUBJECTIVE CRITERIA (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Team Participation On-Site (X1)	The majority of the construction is completed by one member of the team; the partner may be disengaged.	Both team members generally are engaged in the process, though one member may take on more responsibility than the other.	Both team members are actively involved in the construction; there is shared responsibility between team members.	
Pre-Built Drawing and Parts List (X1)	The submitted drawing was incomplete, not accurate, of proper quality, or was not to scale; a complete parts list was not included.	The submitted drawing was complete but lacked clarity, accuracy, or was of poor quality, the parts diagram was not complete or was incorrect.	The submitted drawing was complete, accurate, and to scale; the parts list was complete and accurate.	
Portfolio (X1)	Portfolio is unorganized and/or missing three (3) or more components; leadership and/or 21 st century skills are not evident.	Portfolio includes most components and is generally organized; leadership and/or 21 st century skills are somewhat evident.	All components of the portfolio are included in order, and content and organization are clearly evident; leadership and/or 21 st century skills are clearly evident.	
SUBJECTIVE CRITERIA SUBTOTAL (30 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (70 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				
TOTAL (110 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

